

EMMANUEL YEBOAH

Machine Learning and Data Engineer

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PROFESSIONAL SUMMARY

Data professional with experience using Python and SQL to analyze, clean, and transform complex datasets, automate reporting, and build repeatable data workflows. Hands-on experience developing data pipelines with Apache Airflow and AWS S3, along with machine learning workflows using PyTorch, MLflow, and FastAPI. Strong quantitative foundation in mathematics and physics, with practical experience across data preparation, model training, evaluation, experiment tracking, and deployment.

TECHNICAL SKILLS

Programming: Python, SQL, Bash

Data Engineering: Apache Airflow, ETL, AWS S3, REST APIs, Pandas, PyArrow, PostgreSQL

Machine Learning: PyTorch, scikit-learn, MLflow, model training, model evaluation, anomaly detection

Deployment and Development: FastAPI, Docker, Git, GitHub, Linux, automated testing

Analytics and Visualization: NumPy, Matplotlib, Power BI, Tableau, Excel, statistical analysis, time-series analysis

PROFESSIONAL EXPERIENCE

HardTanh LLC

Data Analyst | September 2023 to Present

- Analyze operational and simulation data with Python and SQL to identify trends, anomalies, and factors affecting performance.
- Build Python workflows to clean, validate, and transform raw data for analysis, reporting, and downstream use.
- Write SQL queries to extract, join, aggregate, and investigate data across relational datasets.
- Automate recurring data preparation and reporting tasks to reduce manual work and improve consistency.
- Investigate data quality issues and unexpected results, trace problems to source data, and validate outputs before they are used in reporting.
- Develop reports and visualizations that make technical findings easier for stakeholders to understand and use.
- Work with cross-functional teams to define metrics, clarify data requirements, and document analytical processes.

T-Mobile, Arch Telecom

Sales Representative | Fitchburg, MA | October 2022 to August 2023

- Reviewed sales, customer, and transaction data to track store performance and identify changes in key metrics.
- Used Excel and internal reporting tools to organize data and support recurring performance reviews.
- Maintained accurate customer and transaction records across CRM and point-of-sale systems.
- Resolved data inconsistencies and record discrepancies to support accurate reporting.
- Used customer and product information to recommend solutions based on individual needs and purchasing patterns.

PROJECTS

Automated Weather Data Pipeline

Apache Airflow, Python, AWS S3, REST API

- Built an Apache Airflow pipeline to collect weather data from an external API on a scheduled basis.
- Cleaned and transformed incoming weather data with Python before storing processed results in an AWS S3 bucket.
- Structured the workflow into separate ingestion, transformation, and storage tasks to improve maintainability.
- Configured task retries and logging in Airflow to support error handling and make failed pipeline runs easier to diagnose.
- Created a repeatable cloud-based workflow for scheduled data ingestion and storage.

Machine Learning Pipeline

PyTorch, MLflow, FastAPI, Python

- Built a machine learning workflow covering data preprocessing, model training, evaluation, experiment tracking, and inference.
- Trained and evaluated a prediction model with PyTorch and used MLflow to record experiments and compare model performance.
- Built a FastAPI endpoint to serve predictions from the trained model.
- Separated preprocessing, training, and inference components to improve reproducibility and maintain consistent data handling.
- Added automated tests and used Git and GitHub for version control and project management.

Time-Series ETL Pipeline

Python, Pandas, PyArrow, Parquet

- Built a multi-stage ETL pipeline to clean, validate, and convert raw CSV data into Parquet for downstream analysis.
- Processed 12 months of time-series data with Python and Pandas to identify trends and changes across reporting periods.
- Organized ingestion, transformation, validation, and analysis into separate stages to make the workflow easier to test and maintain.
- Used structured validation steps to improve data quality before analysis.

EDUCATION

Connecticut College — New London, CT

Bachelor's Degree in Mathematics and Physics

CERTIFICATION

Google Data Analytics Professional Certificate